

Additional materials

1. Porhathai Malaiwong, Allen F. Schroeder, Tia Brown, Chester Wrobel, Nawaphat Malaiwong, Madhumanti Dasgupta, Frank C. Schroeder, and **Michael P. O'Donnell***, Nuclear receptor-neurotransmitter coupling links behavior to metabolic state, *in revision*: Nature (2025). bioRxiv doi: <https://doi.org/10.1101/2025.10.20.683577>

*Corresponding author, project led by O'Donnell lab trainees in collaboration with Frank Schroeder's lab. We identified an intertissue trafficking pathway by which neurotransmitters are intestinally modified by glucosylation and marked with fatty acids indicative of metabolic state. This pathway reconfigures an octopamine signaling circuit to incorporate feeding history via chemical encoding.

2. Christina Campagna, Abigail Descoteaux, Abigail Poole, Eric Peet, Nawaphat Malaiwong, **Michael P. O'Donnell**, and Inna Nechipurenko, Cell specific roles for the conserved galpha chaperone ric8 in cilia biology, *in revision*, MBOC (2026). bioRxiv doi: <https://doi.org/10.64898/2026.02.07.704597>

Collaborative project involving O'Donnell lab and Nechipurenko lab (WPI). Our lab performed calcium imaging and analysis of ciliary mutants, showing that the inversin compartment via G-protein alpha chaperone *ric-8* modulates chemosensory responses.

3. Nawaphat Malaiwong, Porhathai Malaiwong, Chloe Kim, and **Michael P. O'Donnell***, Flint 2.0: Robust and customizable single shot integration in *C. elegans*, *in revision*, G3, (2026). bioRxiv doi: <https://www.biorxiv.org/content/10.64898/2026.01.12.699147v1>

*Corresponding author, project by O'Donnell lab trainees. We developed a highly efficient multicopy transgene method that can be used to generate tunable copy-number integrated transgenes.

4. Rong Xuan Zang, Nawaphat Malaiwong, Ling Wang, Jamie D. Maziarz, Kejue Jia, Bernhard Drotleff, Frank Stein, Mah Noor, C. Jackson Roberts, Mandy Rettel, Jennifer J. Schwarz, Aissam Ikmi, Shigeki Watanabe, Robert Prevedel, Ling Wang, **Michael P. O'Donnell**, Jacob M. Musser, An ancient monoaminergic signaling system coordinates contractility in a nerveless sponge. *bioRxiv* doi: TBD

Collaborative project between O'Donnell lab and Musser lab (Yale). Our lab generated transgenic worms expressing candidate monoamine synthesis and release genes from

sponge. Lead author was co-advised by MOD in this project and O'Donnell lab all *C. elegans* behavioral experiments were implemented in the O'Donnell lab.